

Training Configuration and Summary:

Dataset used: Training: rssi_odom_dataset_12apr25_isolado.parquet | Experiment: rssi_odom_dataset_11apr25_isolado.parquet

model_type: GRU
group_sec: 80
timesteps_orig: 60
batch_size: 256
min_units: 200
max_units: 260
units_step: 20
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['elu']
optimizers: ['rmsprop']
min_layers: 1
max_layers: 1
max_epochs: 50
hyperband_iterations: 3
group_msec: 80
samples_per_sec: 12
forecast_horizon_sec: 3
forecast_steps: 36
timesteps: 60

Dataset Parameters:

Training: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60
Experiment: $\Delta t \approx 12000.0$ ms | samples/window=1.00 | timesteps=60

Tuning:

Trials executed: 4/10

Training Performance:

Loss: 0.0057 | MAE: 0.0576

Validation Performance:

Loss: 0.0085 | MAE: 0.0746

GRU Multi-Step Training Report

Report Generated: 2025-08-05 23:08:35

Training Start: 2025-08-05_22-43-10

Training End: 2025-08-05_22-45-36

Training Duration: 2.43 minutes

Predictions Start: 2025-08-05_22-45-36

Predictions End: 2025-08-05_22-51-07

Predictions Duration: 5.52 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: elu

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 2.2738

MSE: 7.9525

RMSE: 2.8200

R2 Score: 0.6996

Multi-Step Forecast:

Forecast Steps: 36

MAE: 0.0774

MSE: 0.0092

RMSE: 0.0962

R2 Score: 0.7997

Dataset size before down-sample: 40,318 rows
Dataset size after down-sample: 6,059 rows

First 5 timestamps BEFORE:

1744397593300
1744397593310
1744397593332
1744397593362
1744397593375

...

Last 5 timestamps BEFORE:

1744398077967
1744398077978
1744398077991
1744398078003
1744398078016

First 5 timestamps AFTER:

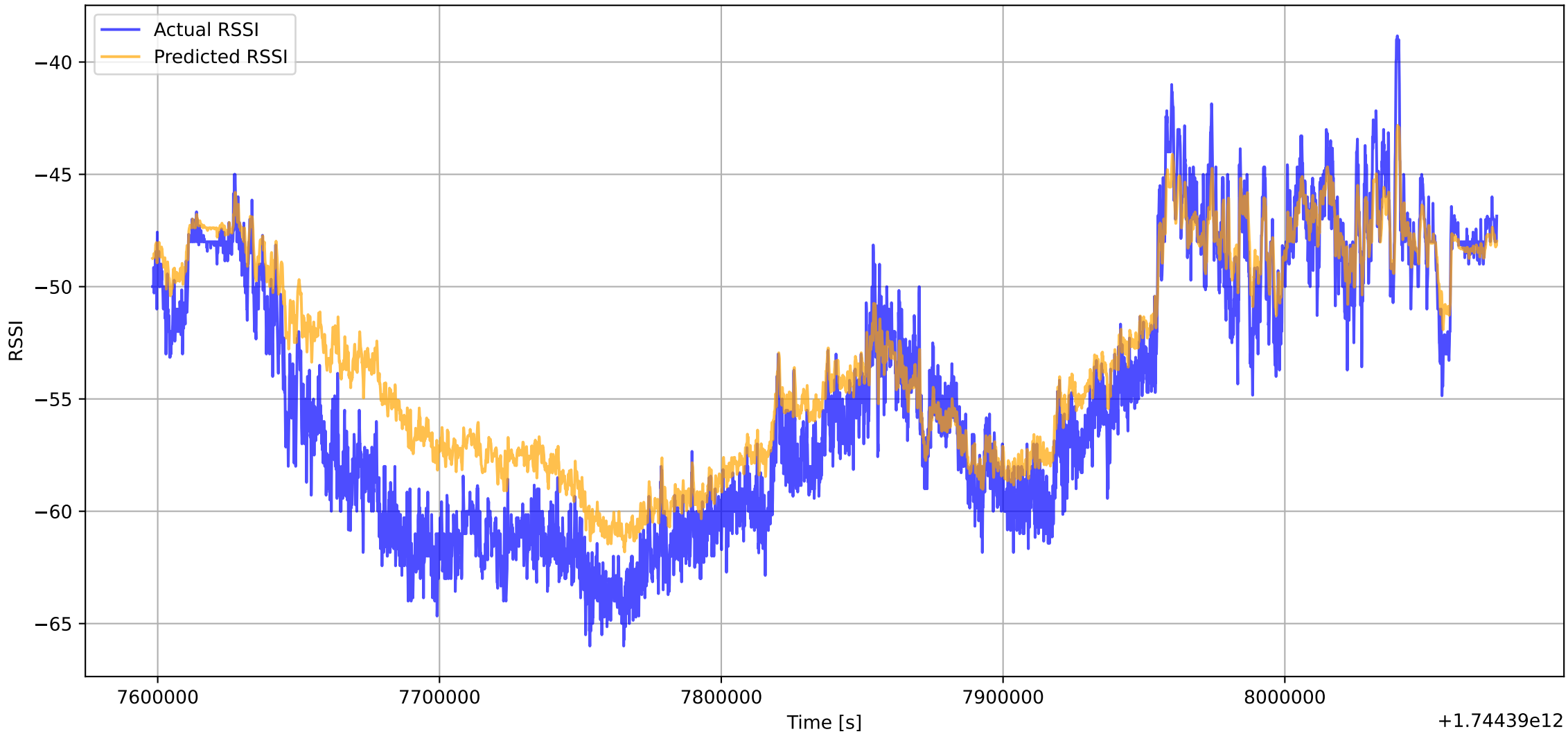
1744397593300
1744397593362
1744397593536
1744397593601
1744397593696

...

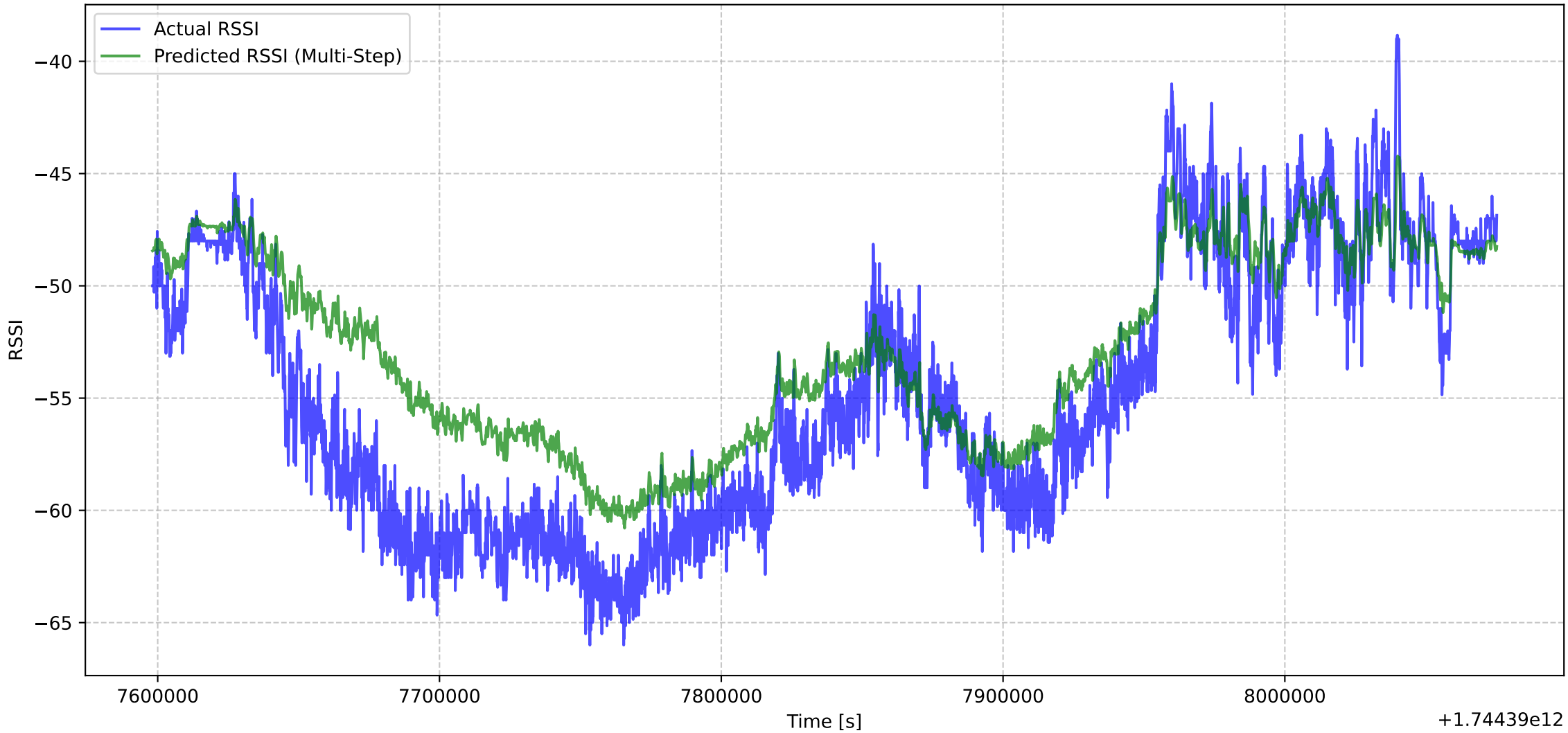
Last 5 timestamps AFTER:

1744398077680
1744398077763
1744398077847
1744398077921
1744398078003

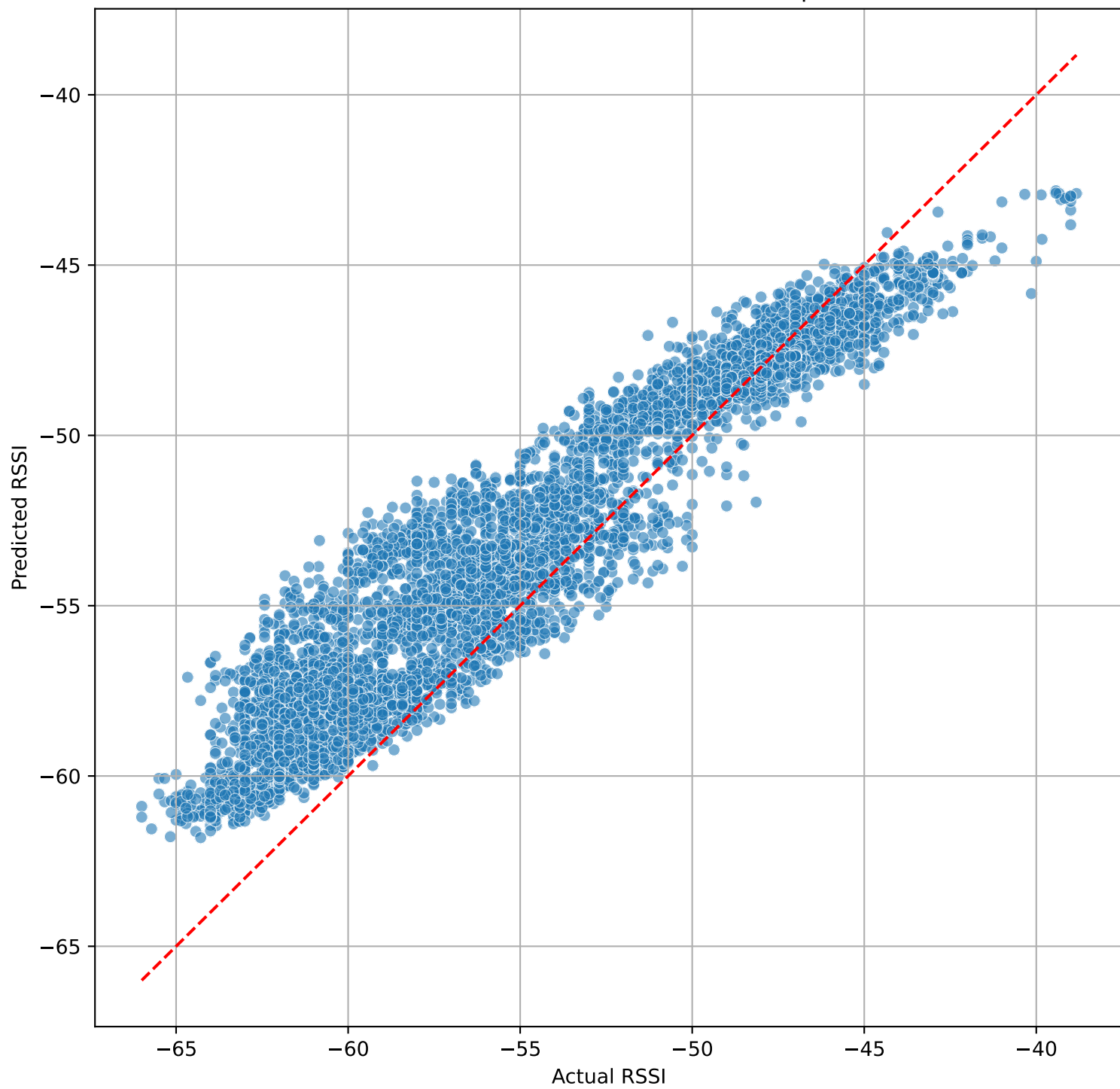
Actual vs Predicted RSSI (First Step)



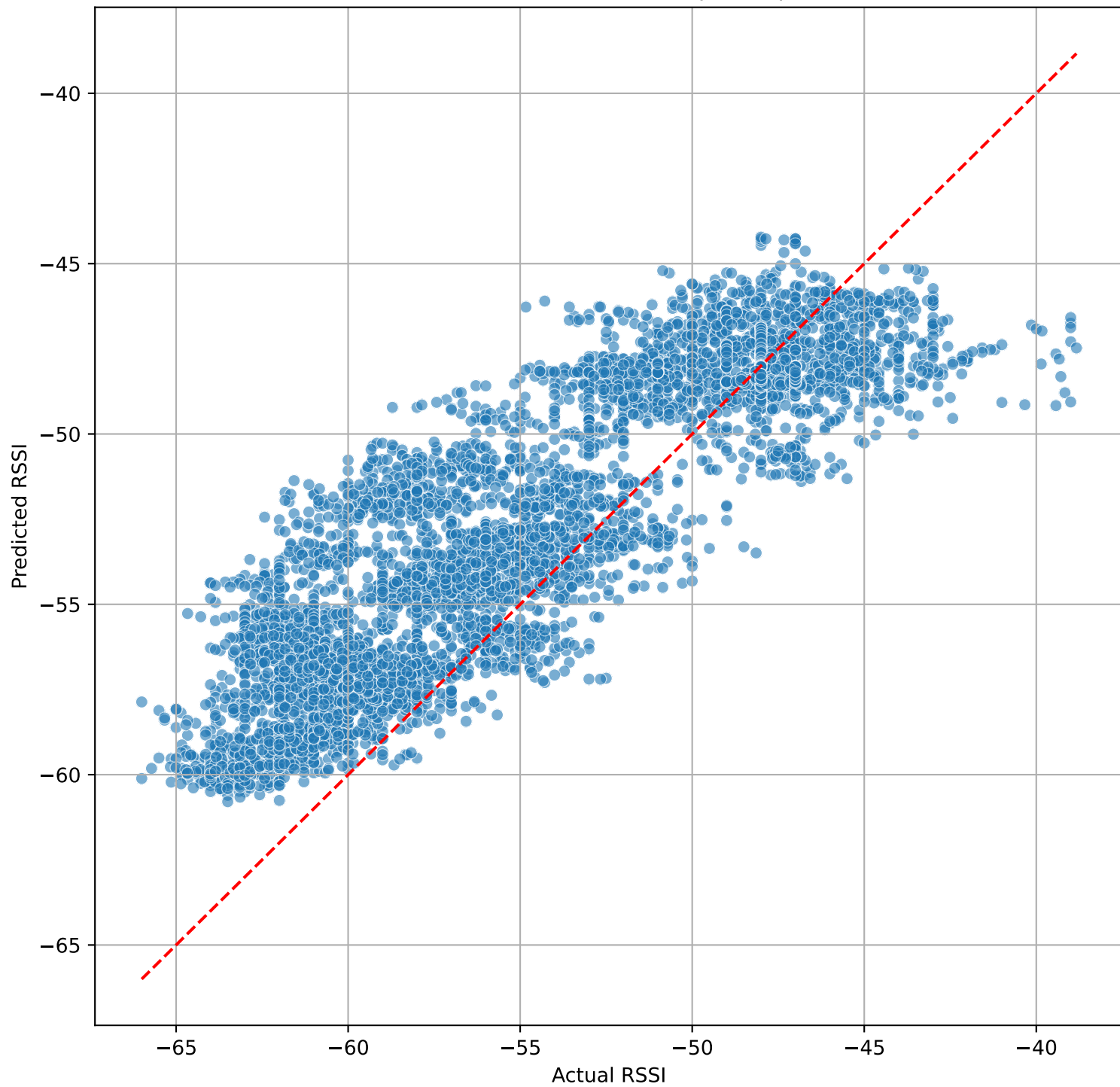
Actual vs Predicted RSSI (Multi-Step (step 36 $\approx$ 3.0 s))



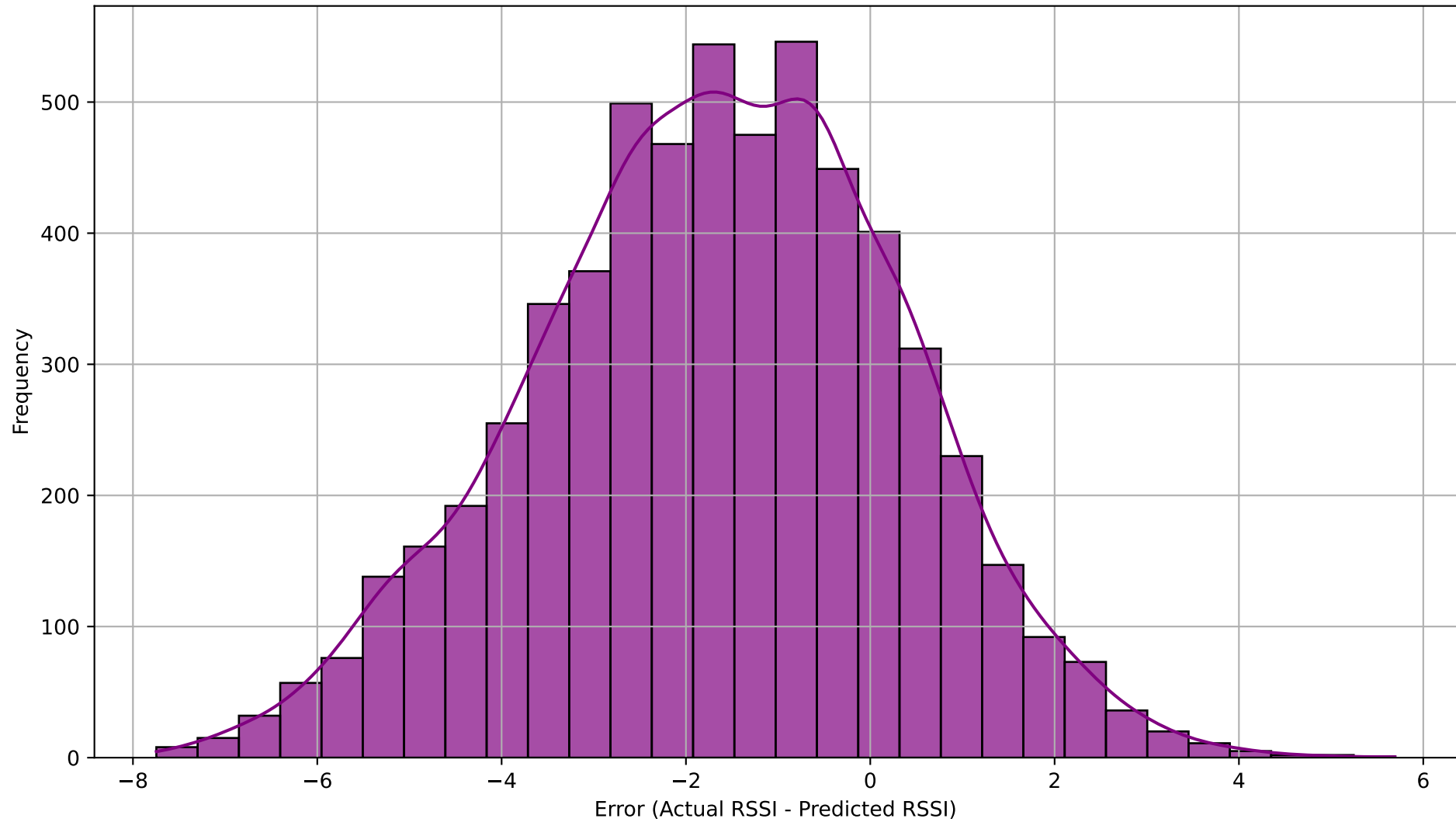
Actual vs Predicted RSSI (First Step)



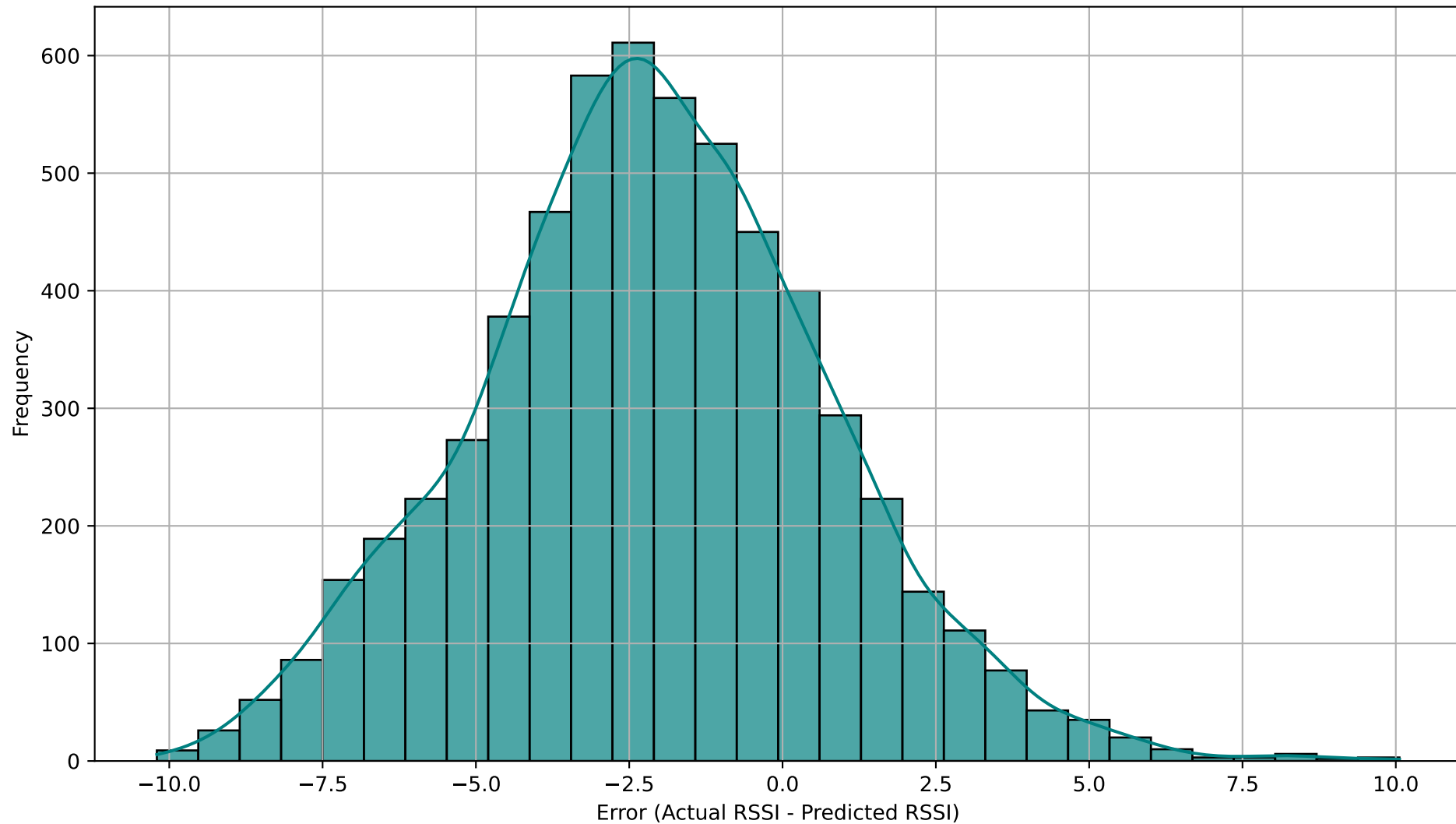
Actual vs Predicted RSSI (Last Step - step 36 $\approx$ 3.0 s)



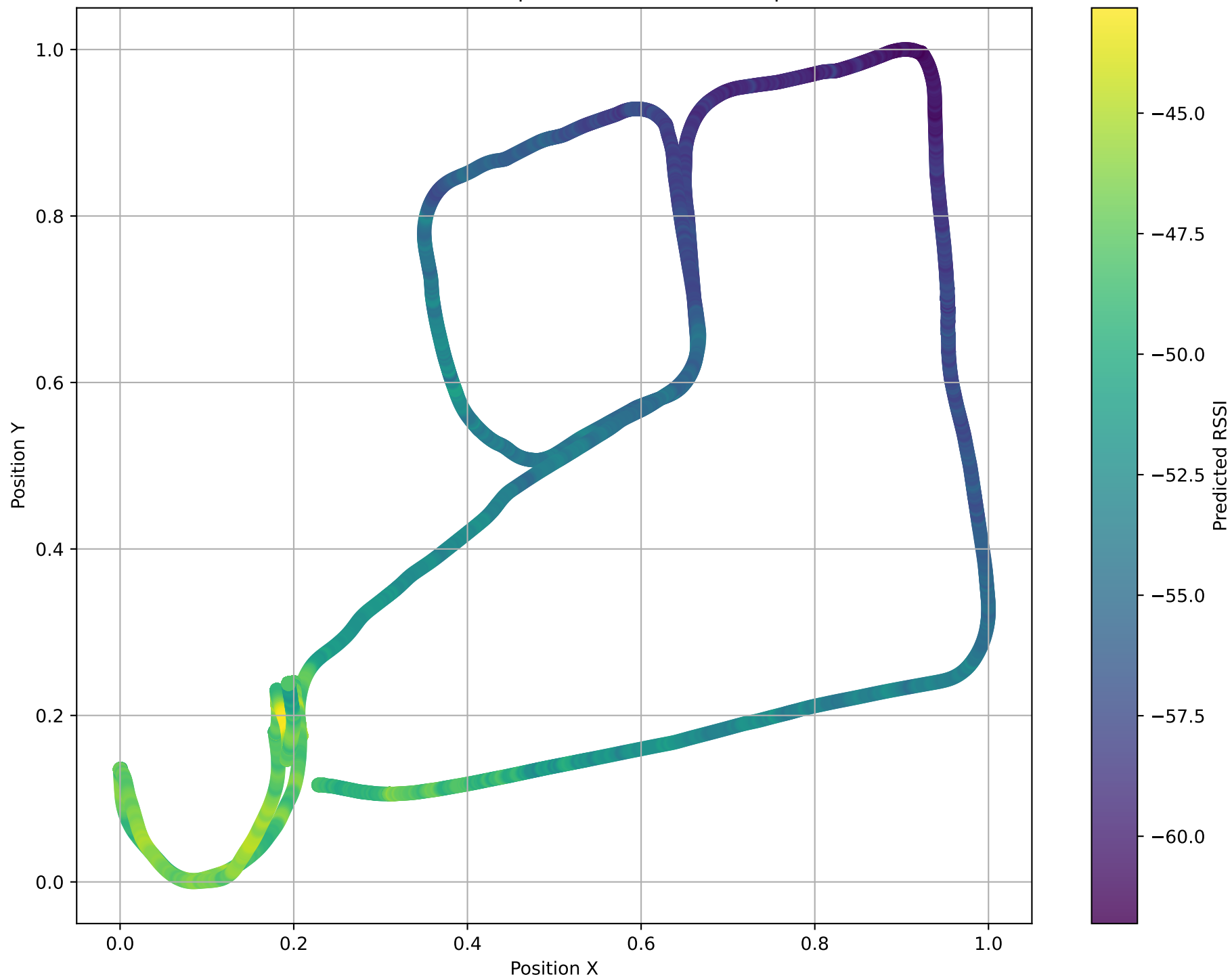
Distribution of Prediction Errors (First Step)



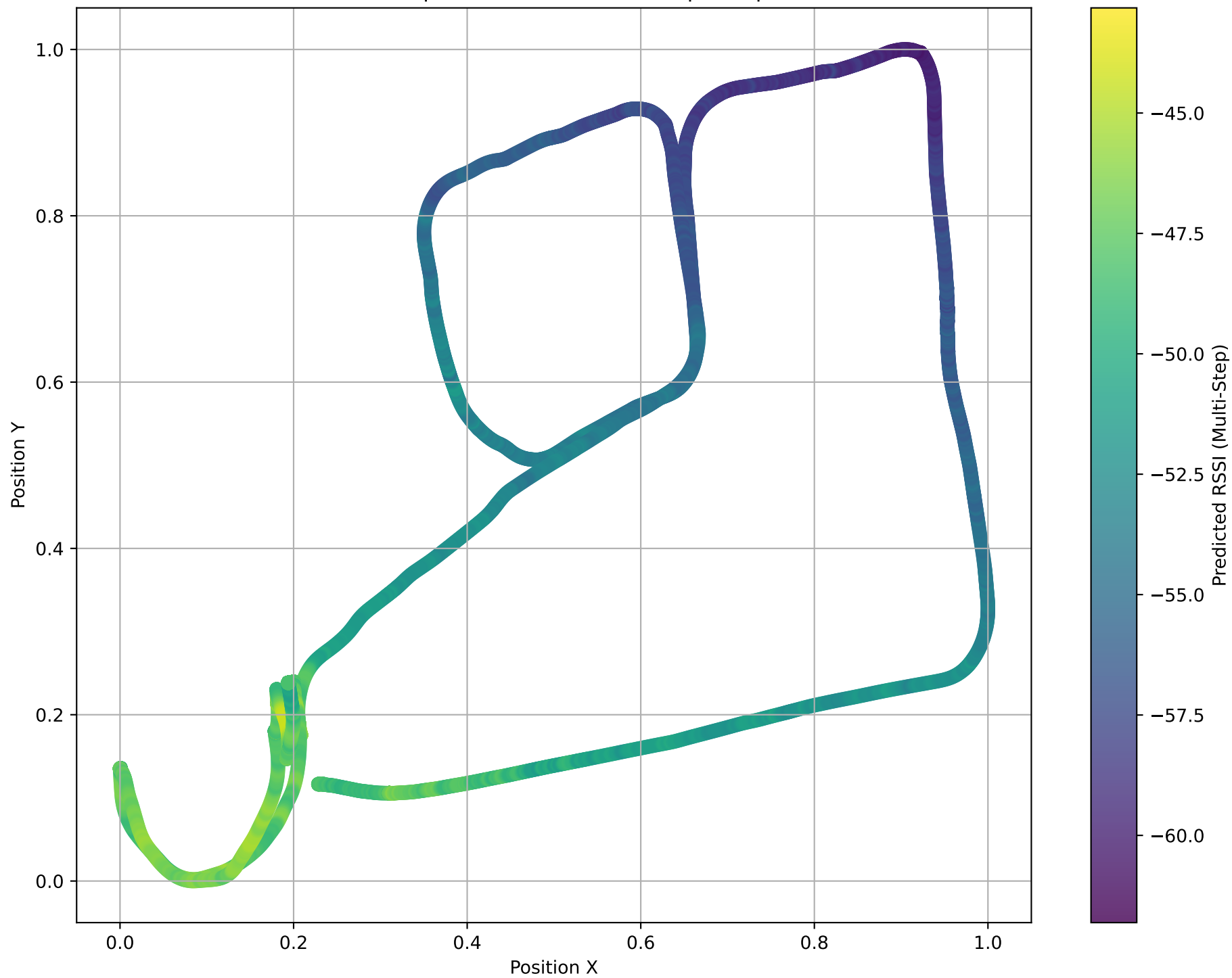
Distribution of Prediction Errors (Last Step - step 36 $\approx$ 3.0 s)



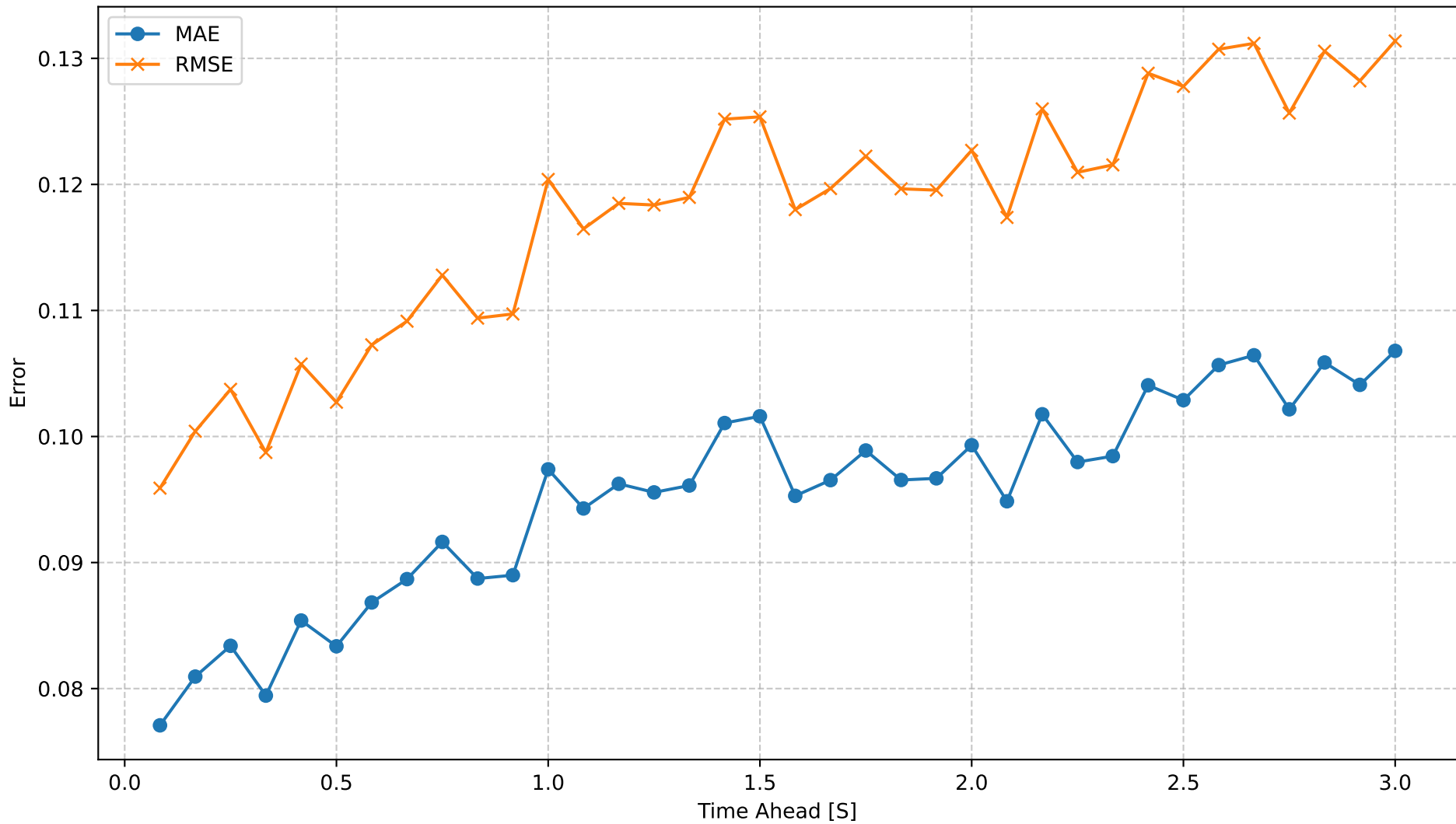
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 36 $\approx$ 3.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: Training: rssi_odom_dataset_12apr25_isolado.parquet |
Experiment: rssi_odom_dataset_11apr25_isolado.parquet
Modelo: GRU
Forecast steps: 36

MAE / RMSE (dB)

- 1º passo : 0.08 / 0.10
- Mediana : 0.10 / 0.12
- 36º passo : 0.11 / 0.13

Crescimento do erro: 38.5% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 36 (0.11), 32 (0.11), 34 (0.11)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.